

22 OHIKOA 2023

$$\boxed{A1} \begin{cases} 3x + y + \alpha z = 0 \\ 2x + \alpha y + z = 1 \\ 3x + \alpha y + z = \alpha - 1 \end{cases}$$

Ebortzi $\alpha = 0$ denean.

$$A' = \left(\begin{array}{ccc|c} 3 & 1 & \alpha & 0 \\ 2 & \alpha & 1 & 1 \\ 3 & \alpha & 1 & \alpha - 1 \end{array} \right) \quad |A| = \begin{vmatrix} 3 & 1 & \alpha \\ 2 & \alpha & 1 \\ 3 & \alpha & 1 \end{vmatrix} =$$
$$= 3\alpha + 3 + 2\alpha^2 - 3\alpha^2 - 2 - 3\alpha =$$
$$= -\alpha^2 + 1$$

$$|A| = 0 \rightarrow -\alpha^2 + 1 = 0 \rightarrow \alpha^2 = 1 \rightarrow \alpha = \pm 1$$

kasuok: $\alpha \neq 1$ eta $\alpha \neq -1$ denean: BATERIAK DETERMINAT (Rouche)
 $|A| \neq 0 \rightarrow \text{heiu}(A') = 3.$
 $\alpha = 1$
 $\alpha = -1$

$\alpha = 1$ denean:

$$A' = \left(\begin{array}{ccc|c} 3 & 1 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 \end{array} \right) \xrightarrow{A} \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} = 3 - 2 = 1 \rightarrow \text{heiu}(A) = 2$$
$$|A| = 0.$$

$$\xrightarrow{A'} \begin{vmatrix} 3 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{vmatrix} = 0 + 3 + 0 - 0 - 0 - 3 = 0.$$
$$\text{heiu}(A') = 2.$$

$\text{heiu}(A) = \text{heiu}(A') = 2$ 81ST BATERIAK INDETERMINAT.

$\alpha = -1$ daveau

$$A' = \left(\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & -1 & 1 & 1 \\ 3 & -1 & 1 & -2 \end{array} \right)$$

$$A \rightarrow \begin{vmatrix} 3 & 1 \\ 2 & -1 \end{vmatrix} = -3 - 2 = -5 \quad \left. \vphantom{\begin{vmatrix} 3 & 1 \\ 2 & -1 \end{vmatrix}} \right\} \text{hein}(A) = 2$$

$|A| = 0.$

$$A' \rightarrow \begin{vmatrix} 3 & 1 & 0 \\ 2 & -1 & 1 \\ 3 & -1 & -2 \end{vmatrix} = 6 + 3 + 0 - 0 + 4 + 3 = 16$$

$\text{hein}(A') = 3.$

$\text{hein}(A) \neq \text{hein}(A')$

SIST. BATERAEZINA

Ebaki: $\alpha = 0$ daveau. $\rightarrow$ SIST. BATERAFARI
DETERMINAVA daveu:

$$\left(\begin{array}{ccc|c} 3 & 1 & 0 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & -1 \end{array} \right)$$

$$x = \frac{\begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & 1 \end{vmatrix}}{1} = -2$$

$$|A| = 1$$

$$y = \frac{\begin{vmatrix} 3 & 0 & 0 \\ 2 & 1 & 1 \\ 3 & -1 & 1 \end{vmatrix}}{1} = 6$$

$$z = \frac{\begin{vmatrix} 3 & 1 & 0 \\ 2 & 0 & 1 \\ 3 & 0 & -1 \end{vmatrix}}{1} = 5$$

$$\boxed{\begin{array}{l} x = -2 \\ y = 6 \\ z = 5 \end{array}}$$